

Pre-screening

Participant	026	
	Yes	No
Do you or have you had a history of migraine headaches?	☐	☒
Do you or have you had a history of claustrophobia?	☐	☒
Do you or have you had a history of frequent or severe motion sickness?	☐	☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐	☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Male
3. How old are you?	46
4. What is the highest level of education you have completed?	☒ High school graduate of GED
5. Which option best describes your current home?	☒ It is rented for money by you or someone in household.
6. How do you pay for healthcare?	☒ Self-pay, out of pocket
8. Which best corresponds to your current work situation?	☒ Working full time.
9. What is your combined family income for the past 12 months (from all sources)	☒ \$20,000- \$49,999

Timeline Followback

Visit 1

Visit 1 Date

10-23-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 2

Visit 2 Date

11-04-2025

Visit 3

Visit 3 Date

11-18-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 4

Visit 4 Date

11-25-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 5

Visit 5 Date

12-01-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Phone

Participant # _____

026

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☐	☒	☐	☐	☐
2. The time I spent in this app just slipped away.	☐	☒	☐	☐	☐
3. I was absorbed in the app task.	☐	☒	☐	☐	☐
4. I felt frustrated while using the app.	☐	☒	☐	☐	☐

5. I found this app to be confusing.	☐	☒	☐	☐	☐
6. Using this app was taxing.	☐	☒	☐	☐	☐
7. The app is attractive.	☐	☐	☐	☒	☐
8. The app is aesthetically appealing.	☐	☐	☐	☒	☐
9. The app appealed to my senses.	☐	☐	☐	☒	☐
10. Using this app was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were videos easy to understand.	☐	☐	☐	☒	☐

AR

Participant ID _____

026

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGROUP Presence Questionnaire

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Disagree |
| 2. I felt that the virtual world surround me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Disagree |
| 4. I did not feel present in the virtual space. | ☒ Completely Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Completely Disagree |
| 6. I felt present in the virtual space | ☒ Completely Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Disagree |
| 11 How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Completely Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☒	☐	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☒	☐	☐	☐	☐
4. I felt frustrated while using the headset.	☐	☐	☐	☒	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☒	☐	☐	☐	☐
8. The augmented reality is aesthetically appealing.	☒	☐	☐	☐	☐

9. The augmented environment
appealed to my senses.



10. Using this system was
worthwhile.



VR

Participant ID _____

026

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGroup Presence Scale

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Somewhat Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space. | ☒ Somewhat Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Somewhat aware |
| 8. I was not aware of my real environment. | ☒ Somewhat Disagree |
| 9. I still paid attention to my real environment. | ☒ Somewhat Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Completely Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Disagree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex.	☐	☐	☐	☒	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☐	☐	☐	☒	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☒	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☐	☐	☐	☒	☐
2. The time I spent in the virtual world just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☐	☒	☐	☐	☐
5. I found this system to be confusing.	☐	☒	☐	☐	☐
6. Using this system was taxing.	☐	☒	☐	☐	☐
7. The virtual environment is attractive	☐	☒	☐	☐	☐

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☒	☐
9. The virtual environment appealed to my senses.	☐	☒	☐	☐	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☒	☐

MR

Participant ID _____

026

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGROUP Presence Questionnaire MR

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Somewhat Agree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Somewhat Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat disagree |
| 6. I felt present in the virtual space. | ☒ Somewhat disagree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat disagree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat disagree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat disagree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☒	☐	☐	☐
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☐	☒	☐	☐	☐
2. The time I spent in mixed reality just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☐	☒	☐	☐	☐
5. I found this system to be confusing.	☐	☒	☐	☐	☐
6. Using this system was taxing.	☐	☒	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The mixed-reality is aesthetically appealing.	☐	☐	☐	☒	☐
9. The blended environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ It is owned or being bought by you or someone in household
2. How do you pay for healthcare?	☒ Self-pay, out of pocket
3. Which best corresponds to your current work situation?	☒ Working full time.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	enjoyed MR the most. notes that it fit right and was much easier to navigate than the other two headsets. felt like VR was okay, but too heavy and was more difficult to navigate. enjoyed AR the least and states that pt did not feel like it was immersive at all nor impactful and states that it was more aggravating to navigate.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	yes, especially MR. other interventions are more person-specific, but believes these technologies have a place in OUD tx and management and notes that they were very peaceful and relaxing for pt.
3. How would you describe the iPAL application, in terms of a cravings management tool?	believes the app was simple and asked appropriate questions. did not feel as if the survey was too long or too short and believes that it is an appropriate cravings management tool.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	HRV - intervention was relaxing and peaceful and they were smooth and easy to do. does not believe the intervention was either too long or too short. CBT - participant did not experience these interventions and they were not recommended for pt, so n/a.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes - feels that the app would be similar to having a sponsor or a helpful outlet. feels like it would be great to have this tool to use on personal time at home to manage cravings, especially when those cravings are high.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes - frequency would depend on how much the pt would feel the need to utilize